Karnataka

Physics- 2016

General Instructions:

1) All the parts are compulsory.

2) Answer without relevant diagram /figure/circuit wherever necessary will not carry any marks.

3) Direct answers to numerical problems without detailed solutions will not carry any marks.

PART-A

I. Answer all the following question:

1. State Faraday's law of electromagnetic induction.

Sol: The magnitude of the induced e.m.f. in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2. Write an expression for the displacement current.

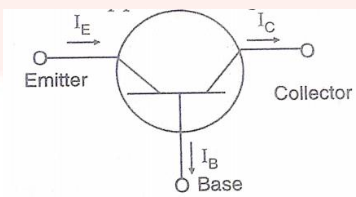
Sol: $I_D = \varepsilon_0 \frac{d\phi_E}{dt}$

3. What is an electric dipole?

Sol: An electric dipole is a pair of two equal and opposite charges separated by a small distance.

4. Draw the circuit symbol of p-n-p transistor.

Sol:



5. How can the resolving power of a telescope be increased?

Sol: By increasing the diameter of the objective. R.P α diameter of Objective.

6. In the following nuclear reaction, Identify the particle X:

$$n \rightarrow p + e^{-} + X$$

Sol: Antineutrino or $\overline{v}$

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7. Define magnetisation of a sample.

Sol: Net magnetic moment per unit volume or Pole strength acquired per unit area of a sample is called magnetisation of the sample.

8. How does the power of a lens vary with its focal length?

Sol: Power of a lens vary inversely with focal length i.e $p\alpha\frac{1}{f}$

9. What is a cyclotron?

Sol: Cyclotron is a machine used to accelerate positively charged particles or ions to high energies.

OR,

Charged particles/ions to high energies.

10 Give the wavelength range of X-ray.

Sol: $10^{-8}m(10nm)$ down to $10^{-13}m(10^{-4}nm)$

II. Answer any five of the following question:

11. The current in a coil of self-inductance 5 mH changes from 2.5 A in 0.01 second. Calculate the value of self-induced e.m.f.

Sol: Inductance of the coil $L=5\times10^{-3}H$

Initial current in the coil $2.5A$

Final current in the coil $2.0A$

Change in current $dI=2.0-2.5=-0.5A$

Time taken for the change,

$dt=0.01$

$$E=-L\frac{dI}{dt}=\frac{-5\times10^{-3}(-0.5)}{0.01}=0.25V$$

12. What is a toroid? Mention an expression for magnetic field at a point inside a toroid.

Sol:

A toroid is a hollow circular ring of finite thickness on which a large number of turns of an insulated wire are closely wound.

$B=\mu_0 nI$

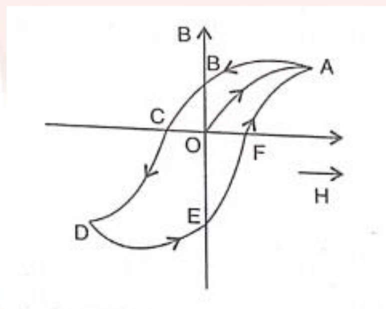
13. What are isotopes and isobars?

Sol: Isotopes of an element are the atoms of the element which have the same atomic number but different atomic weights.

Isobars are the atoms of different elements which have the same mass number but different atomic number.

14. Draw the variation of magnetic field (B) with magnetic intensity (H) when a ferromagnetic material is subjected to a cycle of magnetisation.

Sol:

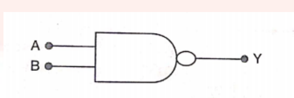


15. Mention two applications of Polaroids.

Sol: (i) Polaroids are used as glass windows in train and aeroplanes.
(ii) Polaroids are used in sun glasses (Goggles).

16. Write the logic symbol and truth table of NAND gate.

Sol:



A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

17. Write two properties of electric field lines.

Sol: (i) Electric lines of force start from a positive charge and end on an negative charge.

(ii) Electric lines of force leave or enter the charged surface normally.

18. What is myopia? How to correct it?

Sol: A human eye is shortsighted (or) myopic, it is can see near objects clearly but is unable to see clearly the far off objects.

To correct short-sighted vision a concave lens is used.

PART-C

lll. Answer any five of the following questions:

19. What is transformer? Mention two sources of energy loss in a transformer.

Sol: The transformer is a device used to raise or lower the voltage of an A.C. supply with a corresponding decrease or increase in current.

Sources of energy losses are

(i) Core or lron

(ii) Copper coil

20. Write three characteristics of nuclear force.

Sol: (i) Nuclear force are strong attractive forces.

(ii) Nuclear forces are charge independent.

(iii) Nuclear forces are short range force.

21. Derive the expression for energy stored in a charged capacitor.

Sol: Let,

$$dU = dW = Vdq = \frac{q}{c}dq$$

The total increase in potential energy in charging the capacitor from $q = 0$ to $q = Q$ is the total energy stored in the capacitor.

$\therefore U = \int dU$

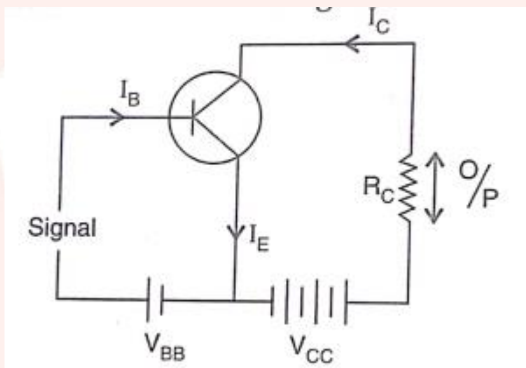
$$= \int_0^Q \frac{q}{c} dq = \frac{1}{c}\left[\frac{q^2}{2}\right]_0^Q = \frac{1}{c}\left[\frac{Q^2}{2}\right] = \frac{Q^2}{2C}$$

$$U = \frac{1}{2}\frac{Q^2}{C}$$

22) What is an amplifier? Draw the simple circuit of transistor amplifier in CE mode.

Sol:

An amplifier is an electronic device that raises the strength of a weak signal.



23) Mention the types of transmission media.

Sol: Transmission media are

(i) Co-axial cable

(ii) Optic fibre

(iii) Free space

24. Derive an expression for drift velocity of free electrons in a conductor.

Sol: If V is the potential difference applied across the end of a conductor of length l, the magnitude of electric field set up is $E = \frac{V}{l}$

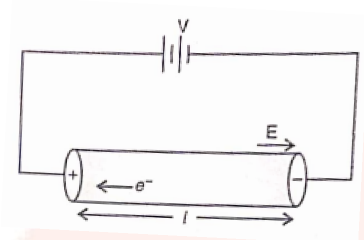
The charge on electron is $-e$,each free electron in the conductor experience force $\bar{F} = -e.\bar{E}$

If m is the mass of an electron, the acceleration of each electron is $\vec{a} = \frac{-e.\vec{E}}{m}$

At any instant of time, the velocity acquired by electron having velocity $\vec{u}_1$ will be

$$\vec{v}_1 = \vec{u}_1 + a\vec{t}_1, \vec{v}_2 = \vec{u}_2 + a\vec{t}_2$$

$\therefore$ The average velocity of all the electrons in the conductor under the effect of external electric field is the drift velocity of the free electrons.



Thus,

$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 .... + \vec{v}_n}{n}$$

$$\vec{v}_d = \frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + ... + \vec{u}_n}{n} + \frac{\vec{a}(t_1 + t_2 + t_3 + ... t_n)}{n}$$

$$\vec{v}_d = 0 + \vec{a}t$$

Where

$$t = \frac{T_1 + T_2 + T_3 .... T_n}{n}$$

$$\therefore \vec{v}_d = at, \vec{v}_d = \frac{-e\vec{E}}{m} T$$

$$\therefore V_d = \frac{eET}{m}$$

25. Explain briefly the coil and magnet experiment to demonstrate electromagnetic induction.

Sol: (i) The arrangement consisted of a coil with a galvanometer and a bar magnet.

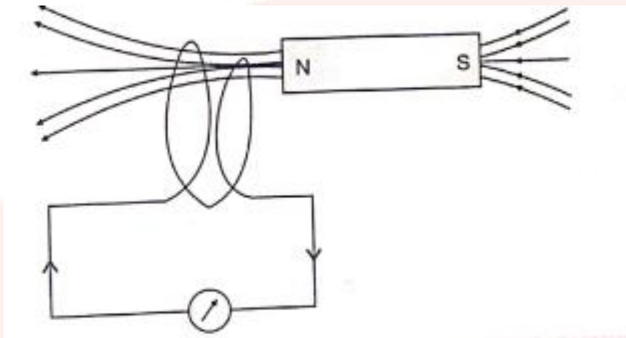
(ii) When North Pole of the bar magnet moved towards the coil, galvanometer should deflect in one direction indicating flow of current in the coil.

(iii) When North pole of the bar magnet moved away from the coil galvanometer again showed the deflection but now in the opposite direction indicating flow of current in the coil but in the opposite direction.

(iv) The deflection of the galvanometer was large when the bar magnet was moved faster towards or away from the coil.

(v) When South Pole of the magnet was brought near the coil, or was moved away from the coil the galvanometer showed deflection in the opposite direction as compared to the available with the similar movement of the North Pole of the magnet.

Conclusion: (i) Whenever there is a relative motion between a closed coil and a magnet, induced e.m.f is set up across the coil or induced current flows through the coil.



26. Write three properties of ferromagnetic materials.

Sol: (i) They are strongly attracted by a magnet.

(ii) Magnetisation (M) has large positive value.

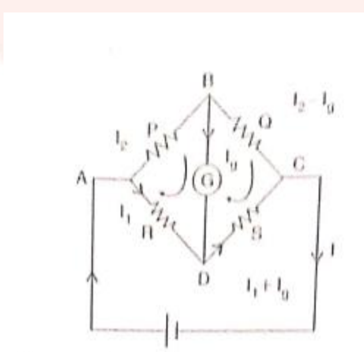
(iii) The sample gets strongly magnetized in the direction of magnetizing field.

PART-D

lV. Answer any two of the following question:

27. Deduce the condition for balance of a Wheatstone's bridge using Kirchhoff's rules.

Sol:



Applying Kirchhoff's loop law to the closed loop ABCD, we get

$$-I_2P - I_gG + I_1R = 0 ------(1)$$

Applying Kirchhoff's loop law to the closed loop BCDB, we get

$$-\left(I_2-I_g\right)Q+\left(I_1-I_g\right)s+I_gG=0------(2)$$

In case of balanced Whetstone Bridge, no current flows through the galvanometer i.e

Hence equation (1) becomes

$$-I_2P+I_1R=0$$
$$I_1R=I_2P$$
$$\frac{I_1}{I_2}=\frac{P}{Q}-------(3)$$

Similarly equation (2) becomes

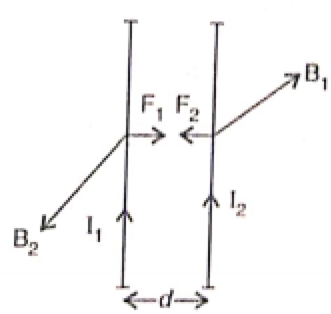
$$-I_2Q+I_1S=0$$
$$I_1S=I_2Q$$
$$\frac{I_1}{I_2}=\frac{Q}{S}----(4)$$

From equation (3) and (4) we get

$$\frac{P}{Q}=\frac{Q}{S}$$
$$or,$$
$$\frac{P}{Q}=\frac{R}{S}$$

28. Derive an expression for the force between two parallel conductors carrying currents. Hence define ampere.

Sol:



Consider two infinitely long parallel conductors carrying currents I_1 and I_2 in the same direction.

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Let ‘d’ be the distance of separation between these two conductors.

The current I_1 in the conductor (1) produces a magnetic field around it as shown in figure. The magnetic field at any point on the conductor (2) due to current I_1 in conductor (1) is given by

$$B_1 = \frac{\mu_0}{4\pi}\left(\frac{2I_1}{d}\right)$$

The direction of B_1 with reference to conductor (2) perpendicular to the plane of the conductor and is directed vertically downward (i.e into the plane)

We know, a current carrying conductor of length ‘l’ placed at right angle to the magnetic field (B) experiences a force. Which is given by

$$F = Bil$$

Therefore, force experienced per unit length of Conductor (2) in the magnetic field B_1 is given by

$$F_2 = B_1 I_2$$

$$= \frac{\mu_0}{4\pi}\left(\frac{2I_1}{d}\right)I_2$$

$$= \frac{\mu_0}{4\pi}\frac{2I_1 I_2}{d}$$

By Applying Fleming’s left hand rule to conductor (2) the direction of F_2 is in the plane of the conductor is directed towards conductor (1)

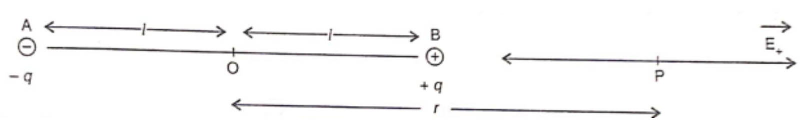
Similarly,

$$F_1 = \frac{\mu_0}{4\pi}\frac{2I_1 I_2}{d}$$

Ampere is that current which if maintained in two infinitely long parallel conductors of negligible cross-section area separated by 1 metre in vacuum causes a force of $2\times10^{-7}\,N$ on each metre of the other wire.

29. Derive an expression for Electric field due to an Electric dipole at a point on the axial line.

Sol:



Consider an electric dipole consisting of $+q$ and $-q$ charges.

$2l$ is the length of the dipole consider a point P at distance r form the center 'O' of the dipole on the axial line of the dipole.

Let a unit positive charge be placed at point P.

Now Electric field intensity at P due to $+q$ charge is given by

$$\vec{E}_{+} = \frac{1}{4\pi\varepsilon_0 (r-l)^2}\hat{i}$$

Electric field intensity at 'P' due –q charge is given by

$$\vec{E}_{-} = \frac{-q}{4\pi\varepsilon_0 (r+l)^2}\hat{i}$$

According to the principle of superposition.

$$\vec{E} = \left(\vec{E}_{+} + \vec{E}_{-}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2}\right]\hat{i}$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{4rl}{\left(r^2-l^2\right)^2}\right]\hat{i}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{2r\left(q\times 2l\right)}{\left(r^2-l^2\right)^2}\hat{i}$$

$$\vec{E} = \frac{2rp}{4\pi\varepsilon_0\left(r^2-l^2\right)^2}\hat{i}$$

V. Answer any two of the following question.

30. Write the experimental observations of Photoelectric effect.

Sol: (i) The process of photoelectric emission is instantaneous.
(ii) The minimum kinetic energy of the photoelectrons increases with the increase in the frequency of incident light, proved the frequency of incident light is greater than the threshold frequency.

(iii) The number of photo electrons emitted per second by a substance is directly proportional to the intensity of incident light greater than the threshold frequency.

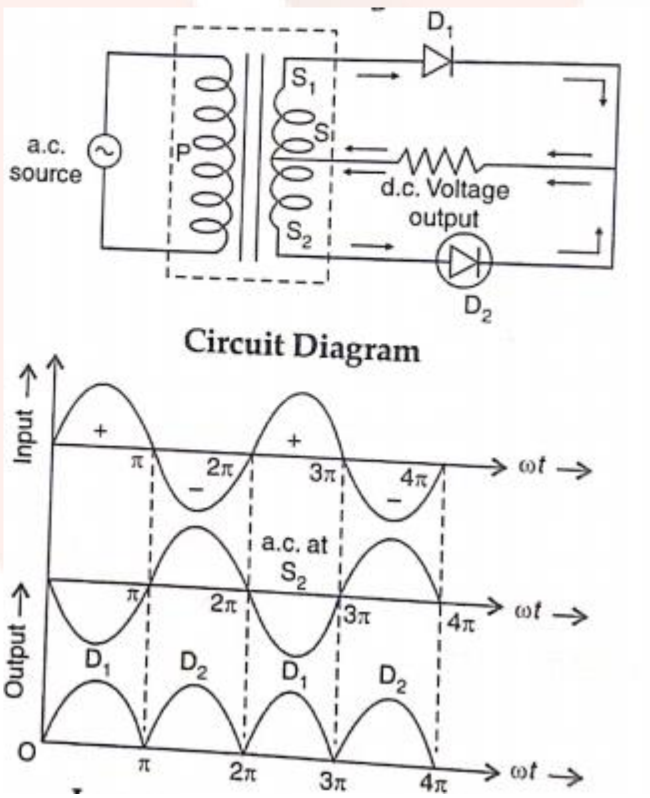
(iv) For a given substance, there is a minimum frequency of incident light called threshold frequency, below which no photoelectric emission takes place.

(v) Photoelectric current is directly proportional to intensity of incident radiation.

31) What is rectification? With relevant circuit diagram and waveforms explain the working of P-N junction diode as a full-wave rectifier.

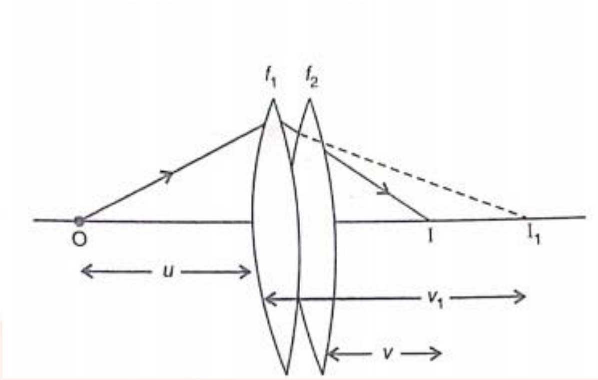
Sol: ‘A’ device which converts alternating current a.c. into direct current d.c. is known as rectifier. The process of converting a.c. in to d.c. is known as rectification.

Working: When positive half cycle of input a.c. signal flows through the primary coil, induced e.m.f is set up in the secondary coil due to mutual induction. The direction of induced e.m.f. is such that the upper end of the secondary coil becomes positive while the lower end becomes negative. Thus diode D_1, is forward biased and diode D_2 is reverse biased, so the current due to diode D_1 flows through the circuit in a direction shown by arrows above (R_L).The output voltage which varies in accordance with. The input half cycle is obtained across the load resistance (R_L). During negative half cycle of input a.c. singal, diode D_1 is reverse biased and diode D_2 is forward biased. The current due to diode D_2 flows through the circuit in a direction shown by arrows (below R_1). The output voltage is obtained across the load resistance (R_L)



32) Derive an expression for equivalent focal length of two thin lenses kept in contact.

Sol:



Consider two thin lenses of focal lengths f_1 and f_2 respectively placed in contact with each other.

Let 'O' be the point object placed on the principle axis of the lense.

If second lens is not present then the first lens forms an image I_1 of the object 'O' at a distance v_1 from it.

$$\therefore -\frac{1}{u}+\frac{1}{v_1}=\frac{1}{f_1}-------(1)$$

Since, second lens in contact with the first, so I_1 acts as an object to the second lens which form's the image I at a distance v from it.

$$\therefore \frac{-1}{v_1}+\frac{1}{v}=\frac{1}{f_2}-----(2)$$

Adding (1) and (2) we get

$$-\frac{1}{u}+\frac{1}{v}=\frac{1}{f_1}+\frac{1}{f_2}$$

Or,

$$-\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Where,

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$$

Thus, the two thin lenses in contact behave as a single lens of focal length f. This single Lens and its focal length f is called equivalent focal length.

Vl. Answer any three of the following question:

33) In Young's double slit experiment, fringes of certain width are produced on the screen Kept at a certain distance from the slits. When the screen is moved away from the slits by 0.1m, fringe width increases by 6×10^{-5}.The separation between the slits is 1mm.Calculate The wavelength of the light used.

Sol:

$$D = 0.1m$$
$$\beta = 6\times10^{-5}m$$
$$d = 1\times10^{-3}m$$
$$\beta = \frac{\lambda D}{d}$$
$$\lambda = \frac{\beta.d}{D} = \frac{6\times10^{-5}\times1\times10^{-3}}{0.1} = \frac{6\times10^{-8}}{0.1}$$
$$= 6\times10^{-7}m$$

34) When two capacitors are connected in series and connected across 4kV line, the energy Stored in the system is 8 j. The same capacitors, if connected in parallel across the same line, the energy stored is 36 J. Find the individual capacitances.

Sol:

$$E_s = \frac{1}{2}C_sV^2 = \frac{1}{2}\left[\frac{c_1c_2}{c_1+c_2}\right]V^2 -----(1)$$
$$E_p = \frac{1}{2}C_pV^2 = \frac{1}{2}[C_1+C_2]V^2 -----(2)$$
$$8 = \frac{1}{2}\left[\frac{C_1C_2}{C_1+C_2}\right]\times\left(4\times10^3\right)^2 -------(3)$$
$$36 = \frac{1}{2}[C_1+C_2]\times\left(4\times10^3\right)-------(4)$$

On solving equation (3) and (4) we get

$$C_1 = 3\mu F, C_2 = 1.5\mu F$$

35) Calculate the shorted and longest wavelengths of Balmer series of hydrogen atom. Given $R = 1.097\times10^{7} m^{-1}$.

Sol: The lines of Balmer series is given by

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

For short wavelength,

$$n = -\infty$$

$$\frac{1}{\lambda} = R\left[\frac{1}{4} - \frac{1}{(\infty)^2}\right]$$

$$\frac{1}{\lambda} = \frac{R}{4}$$

$$R\lambda = 4$$

$$\lambda = \frac{4}{1.097\times10^7}$$

$$\lambda = \frac{4}{1.097}\times10^{-7}$$

$$\lambda = 3646\overset{0}{A}$$

For long wavelength

$$\frac{1}{\lambda_{long}} = R\left[\frac{1}{(2)^2} - \frac{1}{(3)^2}\right]$$

$$\frac{1}{\lambda_{long}} = R\left[\frac{1}{4} - \frac{1}{9}\right]$$

$$\frac{1}{\lambda_{long}} = R\left[\frac{9-4}{36}\right]$$

$$\frac{1}{\lambda_{long}} = R\left[\frac{5}{36}\right]$$

$$\lambda(5R) = 36$$

$$\lambda = \frac{36}{5R}\times10^{-7} = \frac{36}{5\times1.097\times10^{+7}}$$

$$\lambda_{long} = 6.563\times10^{-7} = 6563\overset{0}{A}$$

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36. Calculate the resonant frequency and Q-factor (Quality factor) of a series L-C-R circuit Containing a pure inductor of inductance 4H, capacitor of capacitor of capacitance 27μF And resister of resistance 8.4Ω.

Sol: Here,

$L = 4H, C = 27\times10^{-6}F, R = 8.4-2$

Resonant frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = \frac{1}{2\times3.142\times\sqrt{4\times27\times10^{-6}}}$$

$$f_0 = \frac{1\times10^3}{2\times3.142\times\sqrt{108}} = \frac{1\times10^3}{6.284\times10.4}$$

$$f_0 = \frac{1\times10^3}{65.353} = 15.30Hz$$

Q – Factor

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{8.4}\times\sqrt{\frac{4}{27\times10^{-6}}} = \frac{1}{8.4}\times\frac{2\times10^3}{3\sqrt{3}}$$

$$Q = \frac{2000}{43.65} = 45.81$$

37.(a) Three resistors of resistance $2\Omega, 3\Omega$ and $2\Omega, 3\Omega$ are combined in series. What is the total resistance of the combination?

(b) It this combination is connected to a battery of emf 10V and negligible internal resistance, obtain the potential drop across each resistor.

Sol:

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$R_1 = 2\Omega, V = 10 volts$
$R_2 = 3\Omega$
$R_3 = 4\Omega$
$R_s = R_1 + R_2 + R_3 = 2 + 3 + 4 = 9\Omega$
$But,$
$$I = \frac{V}{R_s} = \frac{10}{9} = 1.1A$$
$\therefore V_1 = IR_1 = 1.1 \times 2 = 2.2 volts$
$V_2 = IR_2 = 1.1 \times 3 = 3.3 volts$
$V_3 = IR_3 = 1.1 \times 4 = 4.4 volts$

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